Altitudes, medians, and bisectors of t[redacted]gles

1

- **a** A median.
- **b** A perpendicular bisector.
- **c** An altitude.
- **d** An altitude.
- **e** An angle bisector.
- **f** An angle bisector, a perpendicular bisector, an altitude, and a median.

2 $x = 46.5$

3 $x = 4$

4 $x = 8.5$

5

- **a** $x = 9$
- **b** $AD = 14$

6

- **a** $x = 5$
- **b** $BD = 12$

7

- **a** $x = 32$
- **b** $BD = 7$

8 The altitude $\overline{MQ}$ to $\triangle RMP$ formed two right triangles which are $\triangle RQM$ and $\triangle MQP$. Note that $\triangle RMP$ is isosceles with $\overline{RM} \cong \overline{MP}$ which are the hypotenuses of $\triangle RQM$ and $\triangle MQP$, respectively. Observe that the altitude $\overline{MQ}$ is a shared leg of two triangles. By Reflexive Property, $\overline{MQ} \cong \overline{MQ}$. By Hypotenuse-Leg (HL), $\triangle RQM \cong \triangle MQP$. With the use of CPCTC, $\angle RMQ \cong \angle QMP$. Consequently, $\overline{MQ}$ is an angle bisector of $\angle RMP$.

Additional questions

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- **a** An altitude is a line segment from the vertex of a triangle that is perpendicular to the opposite side
- **b** It is an Orthocenter. In other words, it is a line segment joining a vertex to the midpoint of

10

a Median

b Altitude

c Neither

d Altitude

e Both median and altitude

f Altitude

11

a Median

b Altitude

c Median and altitude, can also be called a perpendicular bisector

12 An explanation should include the following:

- A proof that the median of an isosceles triangle forms two congruent triangles
- A proof that the two congruent triangles must be right triangles, and therefore form right angles to the base
- A conclusion that the median satisfies the definition of an altitude